

Qatar-1b

R-2783

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_250715 · created 2026-09-04T00:18:07+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460871.9084 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.037
Transit depth (Rp/Rs)^2	3.68 +/- 1.41 [%]
Orbital Inclination (inc)	82.8 +/- 2.29
Airmass coefficient 1 (a1)	0.903 +/- 0.061
Airmass coefficient 2 (a2)	0.083 +/- 0.099
Scatter in the residuals of the lightcurve fit is	6.71 %
Best Comparison Star	#3 - [337, 84]
Optimal Aperture	3.64
Optimal Annulus	12.22
Transit Duration (day)	0.061 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.037 vs 0.1463 (deviation 0.65σ)
Duration fit vs published	0.061 d vs 0.0692 d (deviation 0.23σ)
Mid-time O–C	7.8 min
Depth SNR	2.7
Residual scatter	6.71 %

Light curve

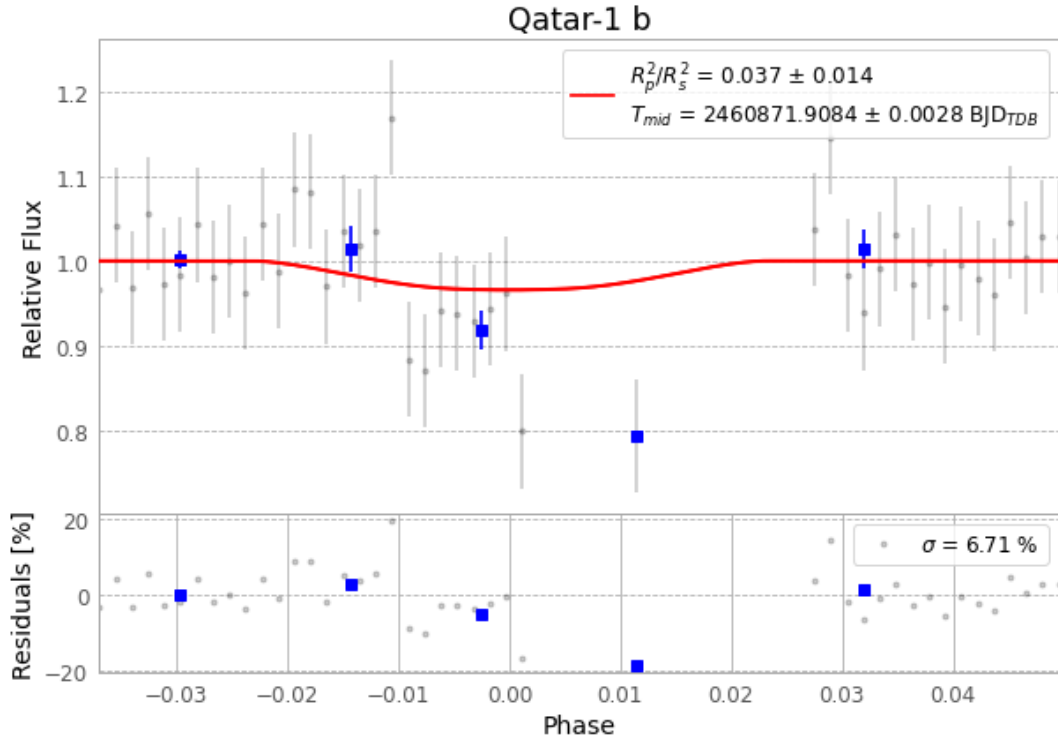


Figure 1 — FinalLightCurve_Qatar-1 b_2025-07-15.png (SHA-256 70076b441f4b3543...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [345, 278], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c630f3dd30d00c4a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

60 FITS frames (39 MB), Qatar-1250715083016.FITS → Qatar-1250715112715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-250715.log (f33388000c5a...), FinalLightCurve_Qatar-1 b_2025-07-15.png (70076b441f4b...), FinalLightCurve_Qatar-1 b_2025-07-15.csv (bc7a907a215a...)

Compute resources

Wall time 232.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 00:18Z.